

SUMIT KUMAR

GenAI & Multi-Agent Engineer | Full Stack Engineer

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Portfolio: my-portfolio-six-gold-86.vercel.app

PROFILE SUMMARY

Full-Stack Engineer with over 3 years of professional experience, backed by a PG Certification in Generative AI and Multi-Agent Systems from IIT Mandi (TIH). Skilled in building scalable RAG architectures, agentic workflows, and LLM orchestration pipelines using LangChain, LangGraph, and CrewAI. Proven track record of integrating AI-driven automation into enterprise MERN/Next.js applications — cutting manual processing time by up to 60% and improving system performance by 15%.

CORE COMPETENCIES

Generative AI & LLMs | Multi-Agent Systems | RAG Pipelines | LangChain | LangGraph | CrewAI | AutoGen | Prompt Engineering | LLM Fine-Tuning | Vector Databases | FastAPI | MERN Stack | Next.js | Python | Node.js | AWS | Azure | Docker | Kubernetes | Azure AI Foundry | Vertex AI | CI/CD | JWT / OAuth2

TECHNICAL SKILLS

- Generative AI & LLMs:** RAG Pipelines, Prompt Engineering (Zero-shot/Few-shot/CoT), LLM Fine-Tuning, Transformers.
- Agentic Frameworks:** Multi-Agent Systems, LangChain, LlamaIndex, LangGraph, CrewAI, AutoGen, ReAct, MCP.
- AI Development:** Gemini, OpenAI, Claude, Hugging Face, FastAPI, RESTful APIs, JWT Authentication.
- Database & Reporting:** Vector DB (Pinecone, ChromaDB, FAISS), MySQL, PostgreSQL, MongoDB, Metabase, SSRS.
- Full Stack Core:** React.js, Next.js, Python, Pandas, TypeScript, JavaScript, Node.js, Express.js, Tailwind CSS, MUI.
- Deployment & LLMOps:** Docker, Kubernetes, AWS, GCP, Azure, Azure AI Foundry, Vertex AI, Jenkins, CI/CD, Vercel, LangSmith/LangFuse.
- Tools & Utilities:** Git/GitHub/GitLab, Postman, Jupyter, Anaconda, Browser Automation (Playwright).

PROFESSIONAL EXPERIENCE

Software Engineer

KENT RO Systems Pvt Ltd (KENT CamEye)

Apr 2024 - Present

- Engineered a multilingual OCR invoice extraction pipeline using AWS Textract and Mistral LLM, reducing manual data entry operations by 60%.
- Engineered a WhatsApp-integrated RAG chatbot using Sarvam LLM, Gupshup webhooks, and the WhatsApp Template API, backed by PostgreSQL to deliver real-time conversational replies to account queries.
- Deployed Claude Sonnet as a fallback LLM within the extraction workflow to accurately parse complex document formats beyond Mistral's capability.
- Optimized backend server logic and implemented NGINX-based load balancing, decreasing system latency by 15% and ensuring high availability.
- Integrated dynamic Zoho Analytics dashboards with role-based and criteria-based filters, enhancing data-driven decision-making and accessibility across departments.
- Established secure authentication and role-based access control using JWT and OAuth2 for internal and external stakeholders.
- Developed and automated complex enterprise reports using SSRS and Metabase, providing real-time data visibility and accelerating cross-departmental decision-making.

Associate Software Developer

Technoldentity

Aug 2023 - Mar 2024

- Revamped the HRMS platform using Next.js and TypeScript, improving UI performance and overall user experience.
- Developed and documented reusable UI components for the Hectare (US Toll Authority) platform using MUI and Storybook, enhancing system maintainability and cross-team design consistency.
- Built and integrated Node.js backend services and REST APIs supporting the HRMS and Hectare platforms, streamlining data flow between frontend and backend systems.
- Transformed raw, unstructured data into curated datasets to fine-tune an open-source LLM for extracting and classifying data from documents — an early foundation for the AI-driven document automation later scaled at KENT RO Systems.
- Automated end-to-end test cases using Cypress and Playwright, reducing manual QA effort and improving testing efficiency.

PROJECTS

Automated Job Application Agent — *Personal Project*

- Architected an agentic job-application system (FastAPI + React) with a CRAG-based matching pipeline — HyDE query expansion, BGE cross-encoder reranking, and MMR deduplication — retrieving over a Qdrant vector store to semantically match resumes to job postings.
- Built a plugin-based automation engine with a central application state machine, Playwright browser automation, and 10+ ATS integration handlers (Greenhouse, Lever, Workday, Ashby, and others), including a dedicated conversational agent for recruiter chatbots and human-in-the-loop review before submission.
- Evolved the pipeline into a LangGraph multi-agent orchestrator, coordinating dedicated discovery, matching, tailoring, and application agents in place of the original sequential flow.
- Built a tiered multi-LLM router (Ollama for lightweight extraction, Gemini Flash/Pro for structured and complex reasoning) and a scheduler for automated recurring job discovery and batch application runs.

Kent Risengine — *Internal*

- Architected high-performance components and validation schemas using React (SSR) and Node.js to ingest data.
- Integrated a payment gateway into the platform to process dealer transactions and order payments.
- Integrated dynamic Zoho Analytics and SSRS dashboards with role-based filters to expedite data-driven decisions.

Cam Attendance — *Internal/Client*

- Maintained and scaled a full-stack attendance platform utilized by thousands of concurrent employees.
- Managed REST APIs and API gateways (Spring Boot) and frontend modules (Angular) to orchestrate real-time HR data tracking and synchronization.
- Integrated automated data pipelines and reporting to sync and validate check-ins, leave logs, and key HR metrics.
- Automated onboarding and HR data migration workflows, significantly reducing manual intervention.

KEY ACHIEVEMENTS

- Engineered an AI-driven, multilingual invoice automation pipeline, successfully cutting manual data entry operations by 60%.
- Boosted system performance and API response times by 15% through optimized server load handling and refined backend logic.
- Built secure Node.js authentication protocols featuring granular role-based access and integrated AWS certificate handling.
- Strengthened overall platform architecture by establishing modular codebases, improving cross-service data flow and scalability.
- Built reusable automation solutions inspired by enterprise RPA practices (similar to UiPath workflows) using browser automation and custom scripting, reducing manual effort and increasing processing accuracy.

EDUCATION

MTech – Computer Science

Monad University

2022 - 2024

B.E. – Mechanical Engineering

RGPV

2012 - 2016

CERTIFICATIONS & TRAINING

PG Certification in Generative AI & Multi-Agent Systems

IIT Mandi - TIH

Dec 2025 - Jul 2026

- Core Curriculum: Mastered advanced Prompt Engineering, NLP fundamentals, and Retrieval-Augmented Generation (RAG) pipelines using Vector Databases.
- Agentic AI & Fine-Tuning: Architected autonomous Multi-Agent Systems (MAS) using LangChain, CrewAI, AutoGen, LlamaIndex and fine-tuned Large Language Models.
- Key Learnings & Deployment: Applied Deep Learning foundations and Transformer architectures to build and deploy scalable GenAI models via FastAPI.
- Deployed FastMCP servers to enable secure, standardized tool-calling for autonomous AI agents.
- Utilized Pandas for complex data preprocessing within Agentic RAG workflows.

Full Stack Web Developer

Newton School

May 2022 - Mar 2023

- Trained in full-stack development (React.js, Next.js, Node.js, Express.js, MongoDB) while building AI-powered web applications and strengthening DSA and problem-solving through coding contests.